🌑 Shadow Science 🌑

Physical Science — Sunshine & Warmth — Unit 2, Lesson 2 of 6

📚 **Grade:** Kindergarten ⏱️ **Duration:** 30–45 minutes 🔬 **Subject:** Physical Science — Sunshine & Warmth
🔢 **Lesson:** 2 of 6

Light travels in straight lines — when something blocks it, a shadow forms!

📋 **NGSS Standard:**
K-PS3-1 — Make observations to determine the effect of sunlight on Earth's surface

🎯 Learning Objective

Students will investigate shadow Science, make observations during the lesson activity, and explain what they learned using evidence from the lesson.

🎯 The Big Idea

Shadows form when an object blocks sunlight. But shadows aren't just interesting to look at — **shadow length tells us how directly the sun's rays are hitting Earth!** A short shadow means the sun is high in the sky and rays hit the ground directly. Direct rays mean more concentrated warmth. A long shadow means the sun is at a low angle and rays are spread out — less warming per spot.

🔬 The Science

Light travels in straight lines. When an object blocks light, it creates a shadow on the other side. The **angle** of the sun changes throughout the day: low in the morning, high at noon, low again in the evening. A high sun creates **short shadows** because the sun's rays hit directly downward — all that energy concentrates on a small area, warming it quickly. A low sun creates **long shadows** because the rays hit at an angle, spreading energy over a much larger area — so each spot gets less warmth. This is why noon is the warmest part of the day, even though the sun has been up for hours!

> 💡 **Key vocabulary:** shadow, sunlight, angle, direct, spread, concentrate, noon, morning, afternoon

📦 Materials

🏠 What You Need

- **Option A (Outdoor):** A sunny outdoor spot + chalk to trace shadows + a small object to cast the shadow (garden stake, ruler, or block) + tape measure or ruler
- **Option B (Indoor):** A flashlight + a small toy or block + white paper + pencil
- Student worksheet (included)
- Pencils and crayons

🔬 Let's Get Started

Step 1 — What Makes a Shadow? (5 min)
In a room, shine a flashlight at a wall and hold your hand in front of the beam. A shadow appears!
Ask: *"What made the shadow? Where did the light go? Why doesn't the shadow appear in front of your hand?"*
Key point: Light travels in straight lines and can't bend around objects. The shadow is where light is blocked.

Step 2 — Change the Angle, Change the Shadow (10 min)
Indoor method: Place a small toy on white paper. Hold a flashlight directly above (straight down) → short shadow. Move flashlight to a low angle (like sunrise) → long shadow! Trace each shadow outline with pencil.

Outdoor method: Observe shadows at morning, noon, and afternoon. At each time, trace the object's shadow with chalk or on paper. Measure the length and record on the worksheet.

💡 **Flashlight tip:** For the "noon" position, hold the flashlight as high as possible directly overhead. For "morning/evening," hold it at about a 30-degree angle from the side. The difference in shadow length will be dramatic!

Step 3 — Record Your Shadows (10 min)
On the worksheet, students draw the object and its shadow for each sun position. They also circle whether the shadow is short, medium, or long. Discuss together which shadow goes with which time of day.

Step 4 — Connect to Warmth (5 min)
Ask: *"When the sun is low in the morning, is the shadow long or short? Is it warm outside yet?"*
Then: *"When the sun is high at noon, is the shadow long or short? Is it warm outside?"*
Help them connect: short shadow = high sun = more direct rays = warmer!

💬 Discussion Question

🗣️ **"Why is it warmer at noon than at 8am even though the sun has been up for hours?"**

> *→ At noon the sun is directly overhead, so its rays hit the ground more directly and concentrate more energy in one spot. In the morning, the sun is at a low angle — the same amount of energy spreads over a much bigger area, so each spot gets less warmth. It's like the difference between shining a flashlight straight down on a table versus shining it at an angle — the angled beam makes a bigger, dimmer patch of light!*

🚀 Extensions

- 📏 **Shadow Diary:** Go outside every hour and measure your shadow (trace it in chalk or use a ruler). Make a chart! When was it shortest?
- 🌍 **Seasons Connection:** In winter, the sun is lower in the sky all day long — that's why it's colder! The same principle as morning vs. noon, just across the whole year.
- 🔦 **Shadow Theater:** Use a flashlight and hand shadows to make animals on the wall. Notice how the shadow size and sharpness change when you move the flashlight!
- 🗺️ **Sundial:** Stick a pencil vertically into a piece of clay on a sheet of paper. Mark the shadow position every hour. You've made a sundial!

👩‍🏫 Parent/Teacher Notes

- **Best time outdoors:** If doing this outside, early morning + noon works beautifully — the shadow change is dramatic. Even 2 hours apart shows a clear difference.
- **Indoor works great:** The flashlight method is very effective and gives you complete control over the "sun angle." Kids usually find this very visual and engaging.
- **Common misconception:** Many kids think it gets warm because "the sun has been heating the air all morning." Gently redirect: it's about the angle of the rays, not just time elapsed.
- **Connect to Lesson 1:** Remind them that in Lesson 1, they found direct sunlight warms things faster. This lesson explains WHY the sun warms more at noon — more direct rays!

🌑 Student Worksheet

Name: _____ Date: _____

☀️ Trace and Draw Your Shadows

Draw the shadow for each sun position. Circle *short*, *medium*, or *long* for each one.

🌅 Morning Sun
(low in the sky)

Shadow is:
short ***medium*** ***long***
(circle one)

☀️ Noon Sun
(high overhead)

Shadow is:
short ***medium*** ***long***
(circle one)

🌇 Afternoon Sun
(low in the sky)

Shadow is:
short ***medium*** ***long***
(circle one)

🌡️ Connect to Warmth

Circle how warm it feels outside for each sun position:

Sun Position	How Warm? (circle one)
🌅 Morning — low sun	❄️ **Cool** 😐 **Warm** 🔥 **Hot**
☀️ Noon — high sun	❄️ **Cool** 😐 **Warm** 🔥 **Hot**
🌇 Afternoon — low sun	❄️ **Cool** 😐 **Warm** 🔥 **Hot**

🔑 PARENT/TEACHER ANSWER KEY — Detach before giving worksheet to students

Shadow Length Answers

- 🌅 **Morning sun (low angle):** *long* shadow — rays hit at a steep angle, energy spreads out
- ☀️ **Noon sun (overhead):** *short* shadow — rays hit nearly straight down, energy concentrated
- 🌇 **Afternoon sun (low angle):** *long* shadow — same as morning, sun is low again

Warmth Connection

- 🌅 Morning → Cool or mild (low sun, long shadows, spread-out energy)
- ☀️ Noon/early afternoon → Warmest (high sun, short shadow, concentrated direct rays; peak temperature is typically 1–3 PM due to heat accumulation in the ground)
- 🌇 Afternoon → Warm but cooling (sun lowering, though ground holds accumulated heat)

💡 Science Explanation for Parents

This lesson introduces the concept of solar angle and insolation (incoming solar radiation intensity). When the sun is directly overhead, its rays strike a smaller surface area, concentrating energy and causing faster warming. At low angles, the same energy spreads over a larger area — less energy per square inch means slower warming. This is the same reason Earth's poles are colder than the equator: the sun's rays always strike the poles at a low angle, even in summer.

✅ What Success Looks Like

- Child can correctly identify which shadow is shortest (noon) and longest (morning/evening)
- Child can connect high sun → short shadow → more warmth
- Child can explain using their own words: "When the sun is up high, it's warmer"

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